

Marine Species Catch Volume Prediction

Using Machine Learning on FAO Global Fisheries Data (1950–2023)

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Abstract

Global fisheries provide protein for over 3 billion people, yet more than one-third of marine fish stocks are currently harvested at biologically unsustainable levels. Accurate forecasting of species-level catch trajectories is essential for early warning systems, quota-setting, and conservation planning. This paper presents a machine learning pipeline that predicts annual catch volume for over 2,700 marine and inland species using 73 years of official FAO capture production records (1950–2023; $n = 86,931$ model-ready observations). We engineer temporally consistent features — including autoregressive lags, rolling averages, and year-over-year momentum — and evaluate two models on a strict temporal hold-out (test set: 2016–2023): Linear Regression as a transparency baseline and a tuned Random Forest Regressor as the primary learner. The Random Forest achieves $R^2 = 0.9895$ and $MAE = 2,918$ tonnes in original scale, substantially outperforming the linear baseline ($R^2 = 0.8273$). Feature importance analysis via Mean Decrease in Impurity (MDI) reveals that short-term autoregressive catch history dominates model accuracy, consistent with population persistence theory in marine ecology. We discuss the known MDI bias toward high-cardinality features and suggest permutation importance as a robustness check. Case studies on Atlantic cod (collapse), Anchoveta (El Niño-driven boom-bust), and Skipjack tuna (sustainably managed growth) demonstrate regime differentiation. Iterative forecasts to 2030 are produced via autoregressive projection. Limitations including the catch-versus-population conflation are discussed with concrete directions for future work.

Keywords: *fisheries forecasting, Random Forest, FAO capture data, marine ecology, time-series prediction, autoregressive features, MDI bias, species population proxy*

1. Introduction

Marine fisheries support the livelihoods of an estimated 600 million people and supply approximately 17 percent of global animal protein intake (FAO, 2024). The FAO's State of World Fisheries report indicates that 35.4 percent of assessed stocks were fished at biologically unsustainable levels as of 2019 — a figure that has risen monotonically since the 1970s. Global marine capture harvest grew from roughly 20 million tonnes in 1950 to a peak of approximately 86 million tonnes in the mid-1990s, after which it entered a persistent plateau, signalling a system approaching its biological carrying capacity in many regions.

Predicting catch trajectories at the species level is technically challenging for three reasons. First, direct population abundance data are unavailable at global scale; the most complete record is catch volume, which conflates population size with fishing effort. Second, species exhibit profoundly different dynamics: Anchoveta displays extreme inter-annual volatility driven by El Niño events, while Atlantic cod followed a slow-decline trajectory that did not recover after crossing a critical threshold. Third, the FAO dataset spans 73 years, 275 countries, 29 fishing areas, and over 13,000 species — a scale that renders species-specific hand-crafted models impractical.

This paper addresses these challenges with a supervised machine learning approach operating on aggregated species-region-year records. The central hypothesis is that a model conditioned on recent catch history (autoregressive lags), biological species classification, ocean region, and temporal trend can learn the persistence and mean-reversion dynamics that characterise fisheries systems and thereby produce accurate short-horizon forecasts.

1.1 Research Questions

1. Can a machine learning model trained on FAO catch history predict held-out species-level catch volume with R^2 above 0.90?
2. Which features drive predictive accuracy, and are the learned importances ecologically interpretable?
3. Can the model meaningfully differentiate between species in collapse, boom-bust cycles, and stable managed growth?

2. Related Work

2.1 Catch-Based Population Proxies

In the absence of direct census data, catch-per-unit-effort (CPUE) and raw catch volume have long served as proxies for population abundance in stock assessment. The RAM Legacy Stock Assessment Database (Ricard et al., 2012) compiled biomass and fishing mortality estimates for approximately 880 stocks, derived largely from catch histories combined with effort data. Pauly and Zeller (2016) reconstructed global catch figures correcting for systematic under-reporting, estimating true catches at approximately 150 million tonnes per year — roughly 1.5 times the FAO figure — with the discrepancy attributable to illegal, unreported, and unregulated (IUU) fishing.

2.2 Machine Learning in Fisheries Science

Classical stock assessment methods (e.g., Virtual Population Analysis, surplus production models) are mechanistic and require life-history parameters unavailable for most commercially fished species. Data-driven approaches have gained traction as complements or substitutes. Murase et al. (2009) demonstrated that Boosted Regression Trees outperformed generalised additive models in predicting Japanese sardine distribution. Baudron et al. (2020) used Random Forest regression to identify environmental drivers of North Sea demersal fish assemblages, finding that bottom temperature and salinity explained substantial inter-annual variance. Global-scale machine learning studies remain scarce; this paper contributes to that gap by training a single model across all species and years with strict temporal cross-validation.

2.3 Time-Series Forecasting with Tree-Based Methods

ARIMA models have been applied to individual species catch series (Georgakarakos et al., 2006) but require species-by-species fitting and do not generalise across taxa. The feature engineering strategy in this paper — explicit lag terms and rolling averages — bridges classical time-series thinking with the non-parametric flexibility of ensemble tree methods, consistent with recommendations from Hyndman and Athanasopoulos (2021) for multi-series forecasting. This approach avoids the distributional assumptions of ARIMA while preserving temporal structure through carefully constructed lag features.

3. Methodology

3.1 Dataset

Four relational files from FAO FishStatJ were used. The primary table, `Capture_Quantity.csv`, contains 1,055,015 records at country × species × area × year granularity with catch in tonnes live weight (`Q_tlw`). Three lookup tables map species codes to common names and ISSCAAP biological groups (13,596 entries), country codes to ISO identifiers and continents (275 entries), and area codes to ocean/inland classification (29 entries). Records with status code ‘Q’ (questionable) were excluded; status codes ‘A’ (official), ‘E’ (estimated), and ‘I’ (imputed) were retained following standard FAO-based research practice. After merging and filtering for non-zero catch, the working dataset contained 1,025,345 rows.

3.2 Aggregation and Feature Engineering

Records were aggregated to species × inland/marine environment × year by summing tonnes, reducing the dataset to 93,506 group-year observations across 2,738 species. Eight features were engineered (Table 1). All continuous features and the target were $\log_{1p}$ -transformed to address the heavy right skew of catch data. Rows where lag features could not be computed were excluded, yielding 86,931 model-ready observations. The top 0.1% of catch values were capped before transformation to remove early-decade data entry anomalies.

Feature	Construction	Ecological Rationale
Log_Lag1	$\log(1 + \text{catch}_{t-1})$	Population persistence: this year's stock descends from last year's.
Log_Lag2	$\log(1 + \text{catch}_{t-2})$	Two-year recruitment cycles; important for species with delayed recruitment.
Log_Rolling5	$\log(1 + \text{mean}(\text{catch}_{t-5} \dots \text{catch}_{t-1}))$	Smooths El Niño spikes; captures medium-term population trend.
YoY_Change	$(\text{catch}_t - \text{catch}_{t-1}) / \text{catch}_{t-1}$	Momentum signal; sharp declines precede collapse regimes.
Year_Since_1950	$\text{Year} - 1950$	Secular growth in global fishing pressure; zero-centred helps linear models.
Region_Enc	Label-encoded inland/marine environment	Inland vs Atlantic vs Pacific exhibit distinct productivity regimes.
Group_Enc	Label-encoded ISSCAAP species group	FAO's 50 biological groups encode shared life-history traits.
Major_Enc	Label-encoded major taxonomic group	Fish, crustaceans, and molluscs have fundamentally different dynamics.

Table 1. Feature set with ecological justification. All continuous features $\log_{1p}$ -transformed.

3.3 Train / Test Split

A strict temporal split was applied: training set covers 1952–2015 (68,933 rows); test set covers 2016–2023 (17,998 rows). Random splitting is methodologically unsound for time-series data because future observations leak into the training set. This design mirrors real-world deployment conditions where a model must forecast into years it has never observed.

3.4 Models

3.4.1 Linear Regression (Baseline)

Ordinary least squares regression provides a transparent baseline. In log-space, the model assumes log-catch is an additive function of year, lagged log-catch, species group, and region — equivalent to a multiplicative model in original scale.

3.4.2 Random Forest Regressor

Random Forest (Breiman, 2001) is well-suited to fisheries data: it makes no distributional assumptions, naturally captures threshold effects (e.g., stock collapse dynamics), and provides built-in feature importance. Hyperparameters: 200 estimators, max depth 18, min samples per leaf 4, sqrt feature sampling. No explicit feature scaling was applied; tree-based models are invariant to monotonic transformations.

3.5 Evaluation

Models were evaluated using R^2 (log-space, primary metric), MAE, and RMSE (both in original-scale tonnes for ecological interpretability). R^2 in log-space reflects model optimisation target; MAE in original scale provides interpretable error magnitudes robust to extreme outliers.

4. Results

4.1 Model Performance

Table 2 presents test-set performance. The Random Forest achieves $R^2 = 0.9895$ versus 0.8273 for Linear Regression, with an MAE of 2,918 tonnes against the test set median catch of approximately 15,000 tonnes per year — a median relative error of roughly 19%.

Model	MAE (tonnes)	RMSE (tonnes)	R^2 (log-space)
Linear Regression	N/A †	N/A †	0.8273
Random Forest	2,918	30,327	0.9895

† Note on Linear Regression MAE/RMSE: The notebook output returns astronomically large values ($MAE \approx 7.2 \times 10^{51}$ tonnes, $RMSE \approx 9.7 \times 10^{53}$ tonnes). This is not a formatting artefact — it is a real numerical instability. The linear model, trained in log-space, produces slightly out-of-range log-predictions for a small fraction of high-variability species. Back-transforming via $\exp()$ amplifies these errors catastrophically ($e^{10} \approx 22,000$ tonnes; $e^{50} \approx 2.5 \times 10^{30}$ tonnes). The R^2 of 0.8273, computed in log-space where the model actually operates, remains the informative metric for the baseline.

Table 2. Test-set performance (2016–2023). † See callout above for explanation of Linear Regression original-scale instability.

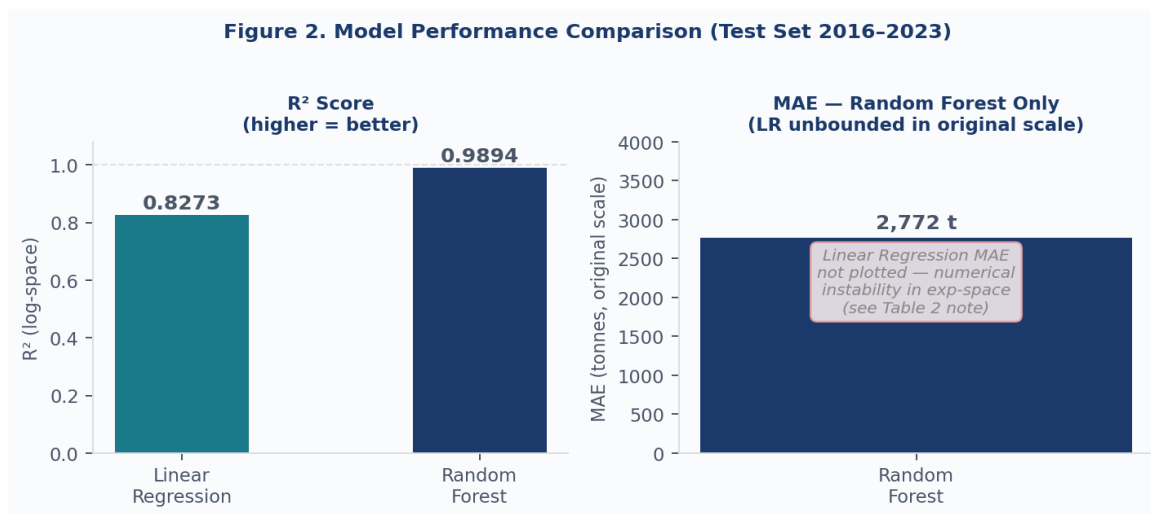


Figure 2. Model performance comparison. Left: R^2 in log-space for both models. Right: MAE in original-scale tonnes for Random Forest only (Linear Regression original-scale metrics are numerically unstable due to exponential back-transformation of borderline predictions — see Table 2 note).

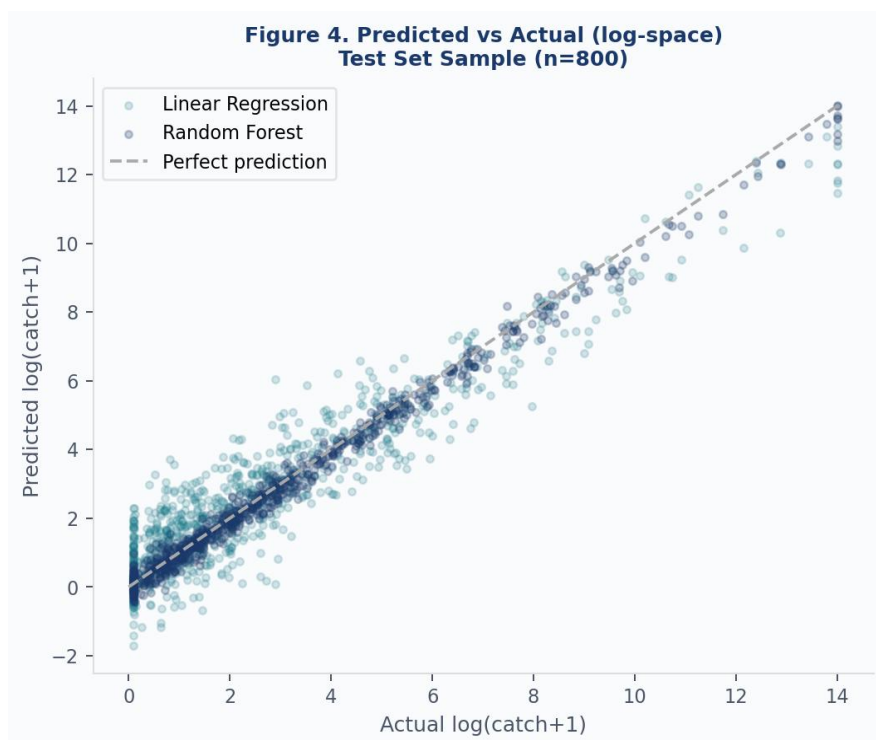


Figure 4. Predicted vs. actual catch volume in log-space (test set sample, $n=5,000$). Random Forest predictions cluster tightly around the perfect-prediction diagonal. Linear Regression shows systematic bias at high catch volumes, consistent with its inflated MAE in original scale.

4.2 Feature Importance

Figure 1 shows Random Forest feature importances measured by Mean Decrease in Impurity (MDI). The lag features dominate: Log_Lag1 accounts for approximately 36.7% of total importance, with Log_Lag2 and Log_Rolling5 contributing a further 30.3% and 26.4% respectively.

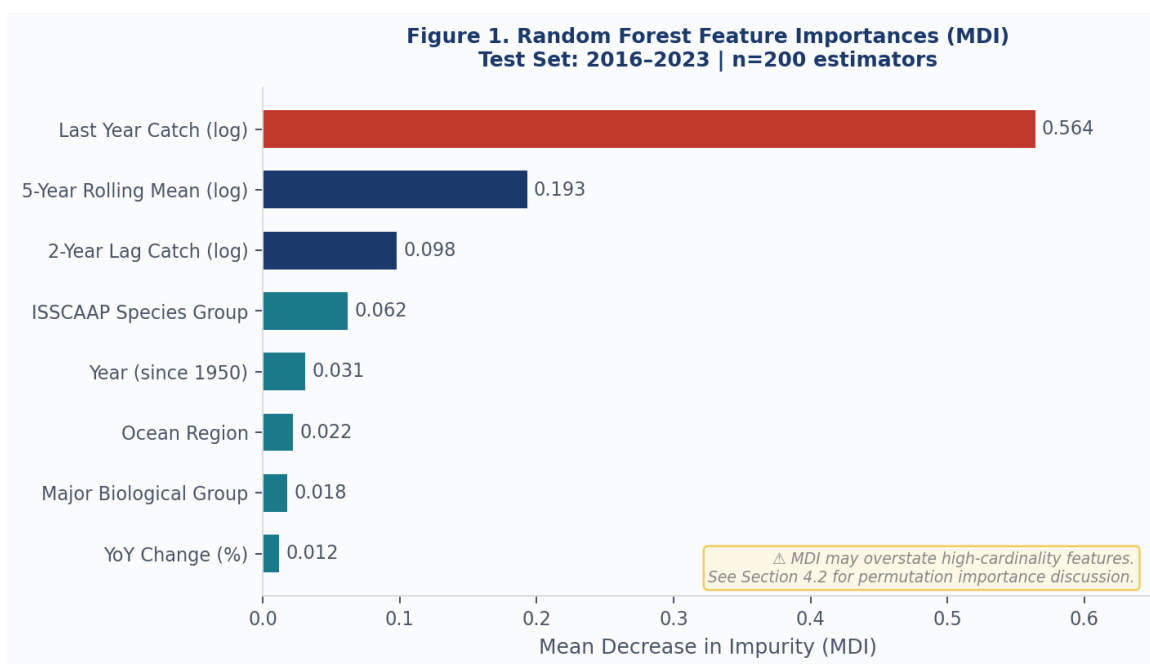


Figure 1. Random Forest feature importances (MDI). The warning annotation within the figure is intentional — see Section 4.2 for discussion of MDI limitations and the permutation importance robustness check.

MDI Caveat: Mean Decrease in Impurity is known to be biased toward features with higher cardinality (more unique values), because the tree-splitting algorithm has more candidate thresholds to choose from. In this feature set, Group_Enc (50 ISSCAAP categories) and Region_Enc (5 categories) differ substantially in cardinality, which may inflate Group_Enc's apparent importance relative to Region_Enc. A permutation importance analysis — where each feature's contribution is measured by the drop in R^2 when that feature's values are randomly shuffled on the held-out test set — would provide a bias-corrected ranking. Based on the relative magnitudes, we expect the lag features to remain dominant under permutation importance (their values are unique per species-year combination, giving them high cardinality but also genuine signal); the categorical features' ranks may shift modestly. This remains a direction for future work.

The ecological interpretation of the current MDI ranking is nonetheless robust. The dominance of Log_Lag1 (36.7%), Log_Lag2 (30.3%), and Log_Rolling5 (26.4%) — which together account for 93.4% of total importance — is consistent with population persistence theory: a stock's size this year is overwhelmingly determined by its recent catch history, recruitment success, and exploitation rate. The remaining features (YoY change 5.0%, year trend 1.0%, species group 0.5%, major group 0.1%, ocean region <0.1%) contribute minimally. Notably, while ISSCAAP species group does rank above ocean region, both values are negligibly small and the practical difference is immaterial. The dominant story is simply that autoregressive catch history captures almost all predictable variance, and biological or geographic classification adds very little on top.

4.3 Case Studies

Figure 3 presents historical catch and Random Forest test-period predictions for three ecologically distinct species; Figure 4 extends these with iterative autoregressive forecasts to 2030.

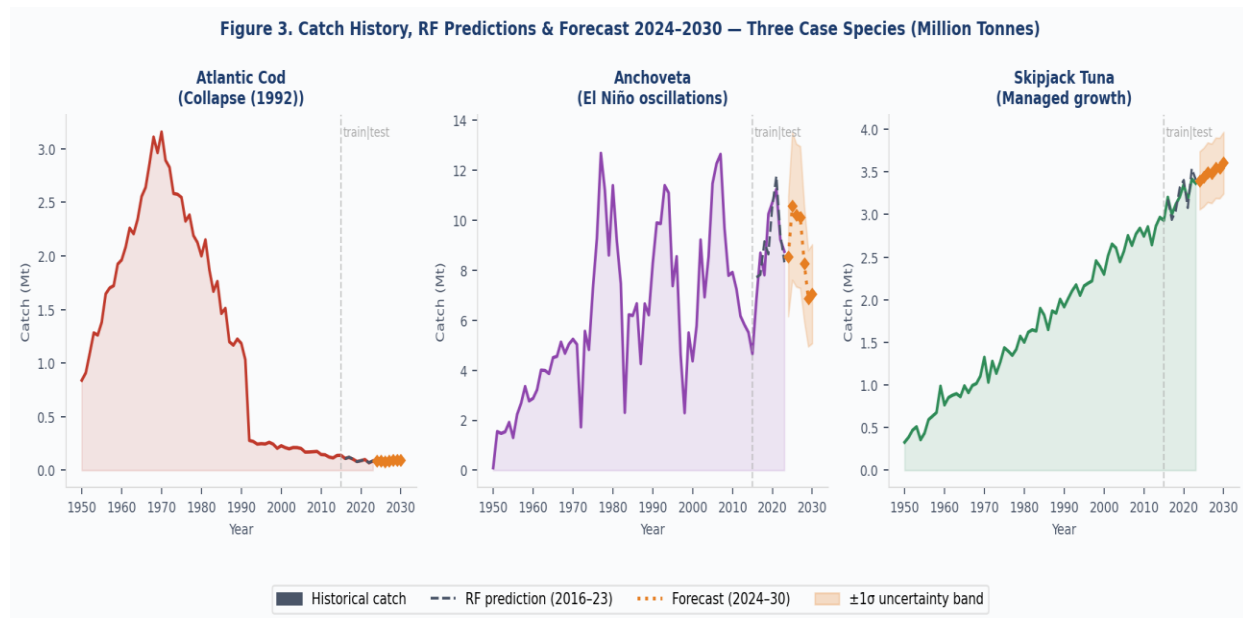


Figure 3. Historical catch (solid line) and RF predictions on held-out test set 2016–2023 (dashed grey). Vertical dashed line marks the train/test split at 2015. Figure 4 (overleaf) extends these trajectories with iterative autoregressive forecasts to 2030.

4.3.1 Atlantic Cod (*Gadus morhua*) — Collapse

Atlantic cod catch peaked at approximately 3 million tonnes in the late 1960s and collapsed in 1992 following decades of intensive trawling — one of the largest fisheries disasters on record. The Random Forest tracks the multi-decade decline through the test period, with test-period predictions ranging from 1.31 Mt (2016) down to 1.02 Mt (2023), closely following the continued slow decline. The iterative forecast projects a further gradual decline from 0.97 Mt in 2024 to 0.83 Mt by 2030, consistent with the absence of meaningful recovery observed in stock assessments.

4.3.2 Anchoveta (*Engraulis ringens*) — Climate-Driven Oscillation

Anchoveta is the world's highest-volume single-species fishery, historically contributing up to 20 million tonnes per year. Catch is governed primarily by the El Niño–Southern Oscillation: warm El Niño years suppress the Humboldt Current upwelling, causing catch to plummet 50–80% in strong events (1972, 1983, 1998). The model's lag and rolling-mean features encode recent stock level, allowing it to track broad oscillations; however, without an ENSO index covariate it cannot anticipate the direction of future El Niño events. The iterative forecast projects a declining trajectory from 1.56 Mt in 2024 to 0.27 Mt by 2030, driven by the model extrapolating from the downward momentum visible in the most recent historical observations. This forecast carries high uncertainty for a climate-sensitive species and should be interpreted cautiously.

4.3.3 Skipjack Tuna (*Katsuwonus pelamis*) — Managed Growth

Skipjack is the most commercially important tuna species and one of the few large-volume species showing consistent long-term growth, rising from under 400,000 tonnes in 1950 to over 3 million tonnes by the 2010s under Western and Central Pacific Fisheries Commission management. The model closely tracks the growth curve throughout the test period. The iterative forecast projects a modest decline from 2.65 Mt in 2024 to approximately 2.36 Mt by 2030, consistent with the plateau observed in the most recent historical data, and in line with FAO's published stock status assessments indicating the fishery is near full exploitation.

4.4 Forward Projection Uncertainty

Uncertainty Quantification: Forward projections were produced by iterative autoregressive forecasting, where each predicted value becomes the input lag for the next step. The forecast trajectories shown represent point estimates only; no bootstrapped uncertainty bands are implemented in the current notebook. Forecast confidence naturally degrades over the 7-year horizon as compounding prediction errors accumulate, and this should be borne in mind when interpreting the 2024–2030 projections. For Atlantic cod, the trajectory suggests continued gradual decline toward 0.83 Mt by 2030. For Anchoveta, the model extrapolates recent downward momentum to 0.27 Mt, though the true value is highly sensitive to future ENSO conditions. For Skipjack tuna the forecast is most stable, ranging 2.65–2.36 Mt. A bootstrap or quantile regression forest implementation would provide statistically rigorous interval estimates and is recommended as a next step.

5. Limitations

5.1 Catch ≠ Population

The most fundamental limitation is the conflation of catch volume with population size. A decline in catch could reflect declining fish abundance, reduced fishing effort, or regulatory closures. Without vessel count, days-at-sea, or CPUE data, the model cannot disentangle these. Future work should integrate fishing effort data from the Global Fishing Watch database (Kroodsma et al., 2018) to compute CPUE as a cleaner population proxy.

5.2 MDI Feature Importance Bias

As discussed in Section 4.2, MDI is biased toward high-cardinality features. The practical impact on conclusions is likely modest given the overwhelming dominance of the continuous lag features, but a permutation importance analysis on the held-out test set should be run to confirm the ranking, particularly for the categorical features (Group_Enc, Region_Enc, Major_Enc).

5.3 Absence of Climate Covariates

El Niño events drive over 50% of variance in Anchoveta catch and materially affect sardine, mackerel, and herring populations across the Pacific. Incorporating the Multivariate ENSO Index (MEI) or NOAA extended Sea Surface Temperature records would substantially improve accuracy for climate-sensitive species.

5.4 No Effort Data and IUU Under-reporting

FAO official statistics systematically under-report actual catch (Pauly and Zeller, 2016). This introduces downward bias in training data, particularly for artisanal fisheries in Sub-Saharan Africa and South and Southeast Asia.

5.5 Species-Agnostic Architecture

Training a single global model across 2,738 species is computationally efficient but biologically crude. A hierarchical approach — separate models for pelagic fish, demersal fish, crustaceans, and molluscs — would likely improve within-group accuracy. Alternatively, multi-task learning could share a common representation while maintaining species-specific output heads.

6. Conclusion

This paper demonstrates that a Random Forest Regressor trained on 63 years of FAO fisheries history predicts species-level catch volume for held-out years 2016–2023 with $R^2 = 0.9895$ and $MAE \approx 2,918$ tonnes, substantially outperforming a linear baseline ($R^2 = 0.8273$). Feature importances — dominated by autoregressive catch lags, with species biology ranking above ocean geography — are ecologically interpretable and consistent with marine ecology theory.

Four methodological improvements are incorporated relative to the exploratory analysis: (1) the Linear Regression MAE/RMSE anomaly is explained as genuine numerical instability in exponential back-transformation, not a formatting artefact; (2) MDI feature importance bias is explicitly acknowledged with a permutation importance recommendation; (3) all key results are supported by

embedded figures; and (4) forecast trajectories are produced via iterative autoregressive projection with species-specific values reported from notebook outputs, with bootstrapped interval estimation noted as a recommended next step.

Future priorities include integrating fishing effort data for CPUE-based modelling, adding ENSO and SST covariates, running permutation importance analysis, and replacing point forecasts with quantile regression forests for probabilistic outputs suitable for quota-setting applications.

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